

Evaluation Report

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Voice Quality

Metric	Method	Result
Latency (TTFT)	Timed across 4 query types	Avg 5.1s, Max 8.2s
STT Accuracy	Deepgram nova-3, en-IN, keyword boosting (Yashraj:5, Guftagu:5)	Domain terms recognized after tuning
Task Completion	E2E booking flow: schedule intent → slot fetch → book	100%

Chat Groundedness

Metric	Method	Result
Hallucination	3 trick questions (fake PhD, fake FAANG, fake awards) + LLM judge	0%
Groundedness	5 queries — every response must trace to a tool call	100%
RAGAS Faithfulness	8 Q&A pairs, GPT-4o-mini judge	0.78
RAGAS Relevancy	Response answers the actual question	0.86
RAGAS Context Precision	Retrieved chunks are relevant	1.00
RAGAS Context Recall	Context contains enough info to answer	0.79

Three Failures and Fixes

- 1. Robot voice — reading a resume out loud.** Agent said *"Experience: He worked at... Skills: Python, C++..."* Sounded like a document, not a person. Fixed by adding conversational style rules to the voice prompt — banned labels, enforced contractions, capped at 2 sentences, added natural transition examples.
- 2. Prompt iteration blind spots.** Went through 4 prompt versions. Each fix broke something else, no way to trace which version caused which regression. Fixed by logging `prompt_version` per request and syncing prompts to Opik's Prompt Library — now every trace shows which version generated it, and I can diff v3 vs v4 in the dashboard.
- 3. LangGraph latency overhead.** TTFT was ~12s. Profiled and found LangGraph's state serialization and graph traversal were the bottleneck — overkill for a linear 4-tool pipeline. Replaced with a 50-line hand-rolled loop. Added semantic cache (cosine sim >= 0.92) for paraphrased queries. TTFT dropped to ~5s, cache hits to ~300ms.

What I'd Improve Next

- **Multilingual voice** — real callers code-switch Hindi/English mid-sentence. Dealt with this in Guftagu (Sarvam AI for Indic), know the hard problems: codeswitching, accents, unnatural Indic TTS.
- **Cost optimization** — ~\$0.02/query is fine for demo, not for scale. Need to experiment with GPT-4o-mini for simple routing, persistent Redis cache, pre-computed answers for top 10 queries.
- **Conversation character** — agent is functional but still feels like AI. Need to work on personality, humor, natural topic transitions, shorter turns with follow-up suggestions.
- **Personalized callers** — recognize returning callers, skip intro. Admin mode from my number for schedule changes.
- **Evals in CI** — run core tests on every PR that touches prompts. Block merge if faithfulness drops below 0.75.